Tanish Rajput

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Education

Noida Institute of Engineering and Technology **2021 – 2025**
B.Tech in Information Technology ***CGPA: 9.01*** *Greater Noida, UP*

Experience

QualtechEdge **Nov 2025 – Present**
AI Engineer *Noida*

- **Engineered FRIDA**, a **config-driven agent platform** that compiles user-designed workflow configurations stored in **PostgreSQL** into **LangGraph agent graphs** at runtime, powering parallel voice agents across loan, insurance, and multiple domains.
- **Built** the agent runtime end-to-end in **Python** and **FastAPI** — input normalization, provider-agnostic **LLM** invocation (**OpenAI**), tool execution via a middleware bridge, and a **Postgres checkpointer** enabling pause/resume sessions across text and voice channels.
- **Shipped** multilingual voice agents on **LiveKit** with **Sarvam AI STT/TTS**, containerized with **Docker** and deployed on **AWS**, instrumented with **OpenTelemetry**, **Langfuse**, and **Prometheus/Grafana**, and evaluated with **DeepEval** for task completion and faithfulness.
- **Developed** supporting agentic systems — **Basil**, a **DeepAgents multi-agent system** generating FRIDA workflows from natural language;**Zeus**, a context-driven **LangChain** variant; and a cross-team **Lip-Sync service** for lip-movement tracking, multi-user presence detection, and fraud signals.

Microsoft **June 2024 – August 2024**
Software Engineer Intern *Noida*

- **Integrated** an **audio panel** into the **Microsoft Designer** editor using **React** and **FluentUI v9,** enabling **audio content creation** and **MP4 export** functionality.
- **Redesigned** the panel UI based on **UX feedback** reducing **user friction** by **20%** and improving overall **usability.**
- **Collaborated** with **Design** and **Accessibility (A11y)** teams to enhance **interface performance** and ensure **WCAG-compliant accessibility.**

Projects

ARGUS

- **Architected** an AI agent authorization gateway using **Gemini Flash/Pro**, **FastAPI**, and **React** that intercepts LLM tool calls and enforces intent-based policies pre-execution — preventing prompt injection and PHI exfiltration.
- **Built** a multi-layer AI pipeline (Intent Extraction → Policy → Analysis) with **Lobster Trap** DPI, **async SQLAlchemy**, **Redis** caching, and SSE live monitoring — processing actions in under 50ms.
- **Designed** a human-in-the-loop review queue with compliance export (CSV/JSON/PDF), backed by **136 tests** and **HIPAA-ready audit trails**

AuthoGraph AI

- **Built** a **multi-agent AI system** using **LangGraph** with a Plan–Execute–Distribute workflow and a **RAG pipeline** powered by **Tavily Search**, vector retrieval, and **LangSmith** monitoring — autonomously researching and publishing technical blogs across Dev.to, Hashnode, LinkedIn, and Medium in under 5 minutes with 95%+ factual accuracy via automated LLM-based evaluation.
- **Developed** a full-stack platform (**FastAPI + React**) with SSE-streamed agent reasoning, **GPT-4o-mini** and **Gemini 2.5 Flash** integration, deployed on **AWS EC2** with a credit-based monetization system via **Razorpay**.

AI & Technical Skills

Programming: Python, Java
Agents & Orchestration: Multi-Agent Systems, LangGraph, LangChain, DeepAgents, Agentic RAG, Context Engineering, MCP, Harness Engineering
Voice AI: LiveKit, Pipecat, Sarvam AI (Indic STT/TTS), Deepgram, ElevenLabs
Evals & Observability: DeepEval, Langfuse, LangSmith, OpenTelemetry, LLM Evaluation
Backend & Data: FastAPI, Celery, Async Python, PostgreSQL (pgvector), Redis, Pinecone, Weaviate, Docker
AI-Native Tooling: Claude Code, Codex, Devin, Cursor, n8n